

Computer Networks

Lecture 1: Introduction

Content



- **General information**
- **Course weightage**
- **Course content**
- **Introduction to Computer Networks**

Regarding Course

- Course Assignment and Lab Submissions: Mahindra *Euclid*
- Keeping the class strength in mind, you are encouraged to contact via email as it may not be feasible to meet in person most of the time.
- It is mandatory to attend all the exams and submit the assignments on time.
- In a supplementary no grade better than grade “D” can be obtained.

Course Content

Introduction: Overview of networking using the Internet as an example, history and development of computer networks, networks topologies, LANs and WANs, OSI reference model, Internet TCP/IP Protocol Stack. Client/server paradigm, End-to-end communication: packet switching and circuit switching

Link Layer: Link layer services, error detection and correction, Sliding Window, Stop and Wait protocols. MAC Layer: Aloha, CSMA, CSMA/CD, CSMA/CA protocols. Examples: Ethernet, including Gigabit Ethernet and WiFi (802.11). Token Ring and to Bluetooth, WiMax

Network layer: Network layer services, IP packet switching, IP addresses- Subnetting, Classless addressing, Network Address Translation, Internet Protocol, IPv6, ARP, DHCP, ICMP, Routing algorithms: Distance vector, Link state, Metrics, RIP, OSPF, BGP, multicast, Inter-domain routing congestion control algorithms

Transport layer: Ports, UDP, principles of reliable data transfer, TCP (handshake, windowing, congestion control) Connection establishment and termination, TCP variants, Use of TCP/IP protocol suite as running example, securing

Application layer: Service requirements, WWW, HTTP, electronic mail, Domain Name System, P2P, socket programming using C.

Text Books

Text Books:

1. Kurose, J and Ross, K Computer Networking: A Top-Down Approach Addison-Wesley- 6th edition, 2012.

Reference Books:

2. Comer, DE (1995) Internetworking with TCP/IP vol I (3rd edition) Prentice Hall ,Most recent edition is 5th edition, (2006).
3. Behrouz A. Forouzan, “Data Communication and Networking”, TMH, 5th Edition ,(2012).
4. Andrew S. Tanenbaum, David J. Wetheral “Computer Networks” Pearson, 5th –Edition-(2011).
5. W. Richard Stevens and G. Gabrani, TCP/IP Illustrated, Volume 1: The Protocols and Unix Network Programming, Volume 1: The Sockets Networking API ,3rd Edition, (2001).
6. Ying-Dar Lin, Ren-Hung Hwang, Fred Baker, “Computer Networks: An Open Source Approach”, McGraw Hill Publisher, (2011)

Course Weightage

<i>Evaluation Parameter</i>	<i>Weightage</i>
Minor I	20%
Minor II	20%
End Sem	40%
Lab Assesment	20%

Introduction

Core foundations of computer networking:

- Data communications
- Networks
- Internet
- Protocols
- Standards



Data Communication

Definition

Data communications are the exchange of data between two devices via some form of transmission medium.

The effectiveness of a data communications system depends on four fundamental characteristics

- **Delivery**. The system must deliver data to the correct destination.
- **Accuracy**. The system must deliver the data accurately.
- **Timeliness**. The system must deliver data in a timely manner. Data delivered late are useless.
- **Jitter**. Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets.

Data Communication

Components of Communication

I. Message.

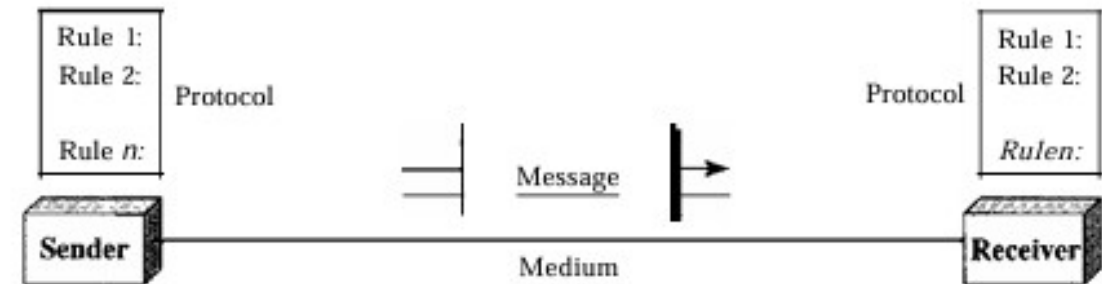
- The message is the information (data) to be communicated.
- Popular forms of information include text, numbers, pictures, audio, and video.

II. Sender.

- The sender is the device that sends the data message.
- It can be a computer, workstation, telephone handset, video camera, and so on.

III. Receiver.

- The receiver is the device that receives the message.
- It can be a computer, workstation, telephone handset, television, and so on.



Data Communication

Components of Communication

IV. Transmission medium.

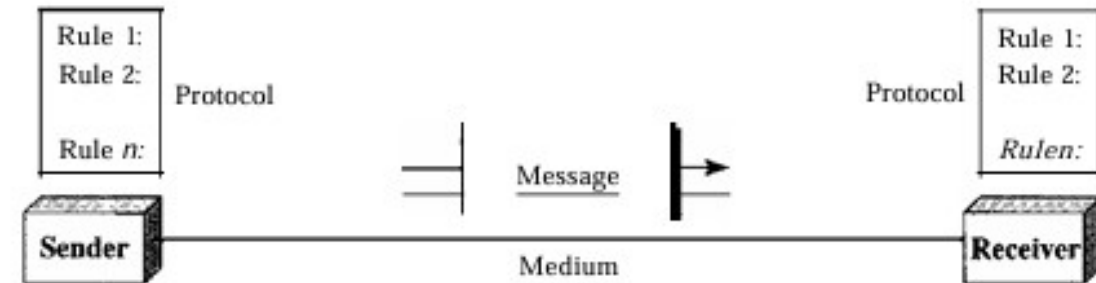
- The transmission medium is the physical path by which a message travels from sender to receiver.

- Wired(Guided) media

Examples :Ethernet cables, Fiber optic cables, Coaxial cables

- Wireless (Unguided) media

Examples :Radio waves (Wi-Fi, Bluetooth), Microwaves (Satellite communication), Infrared (Short-Range Communication)



V. Protocol.

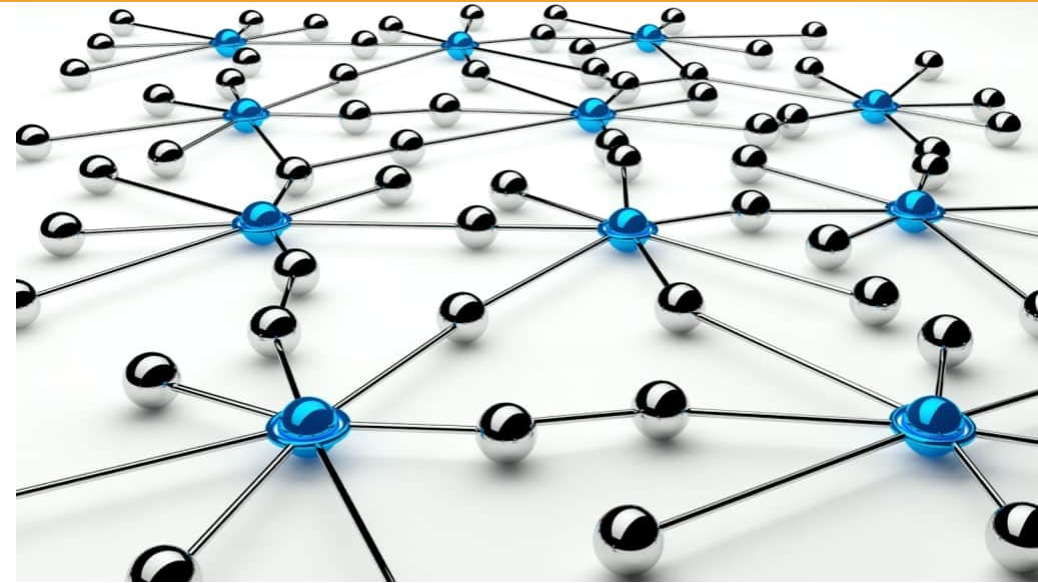
- A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices.
- It determines what data is communicated, how it is communicated, and when it is communicated.

What is network?

Definition

A network is an interconnected system that connects multiple devices (devices are often referred to as nodes) through transmission mediums to share information and resources.

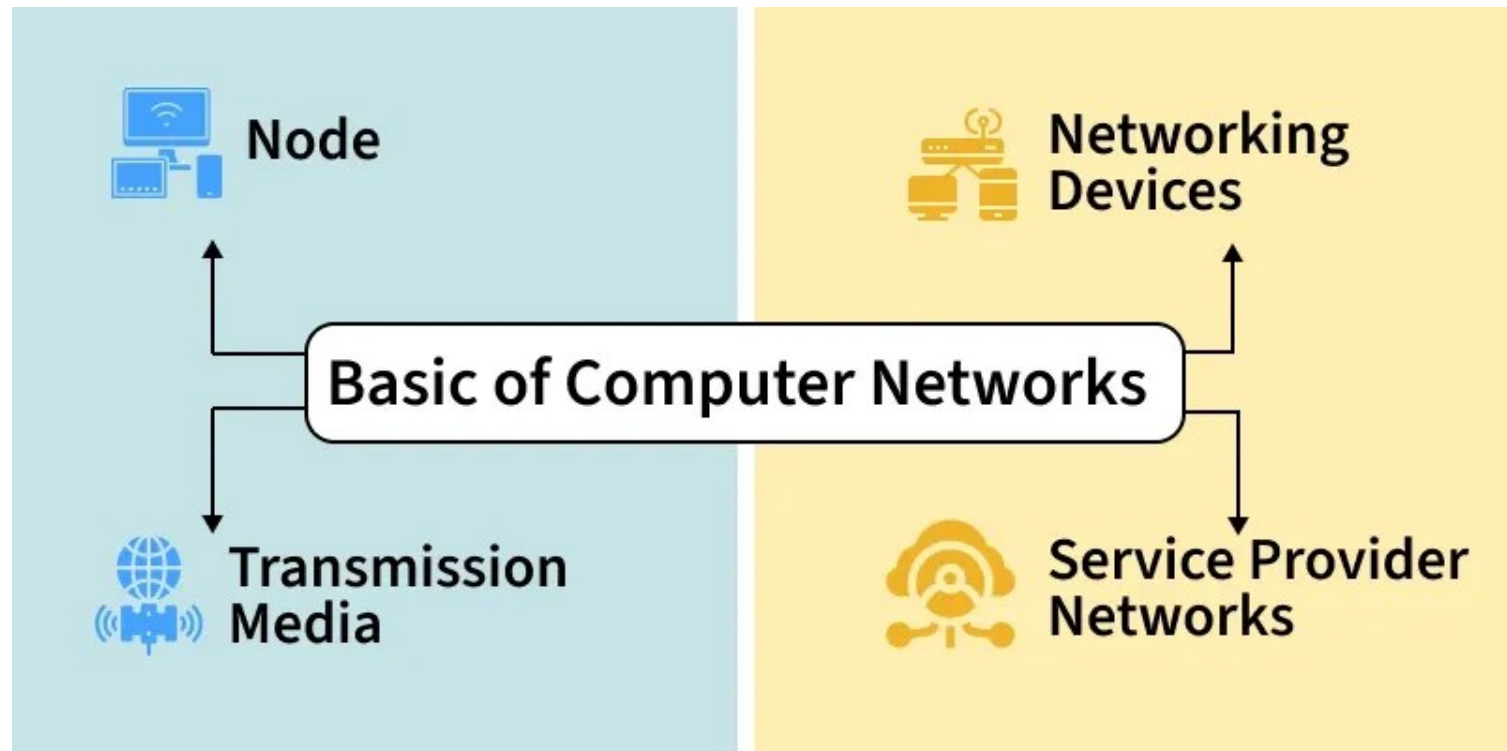
- A transmission medium can be wired or wireless and is used to transport signals carrying information.
- A node can be a computer, smartphone, router, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.



Computer Network

Definition

A computer network is a system of interconnected computers and other devices that can communicate with each other and share resources.

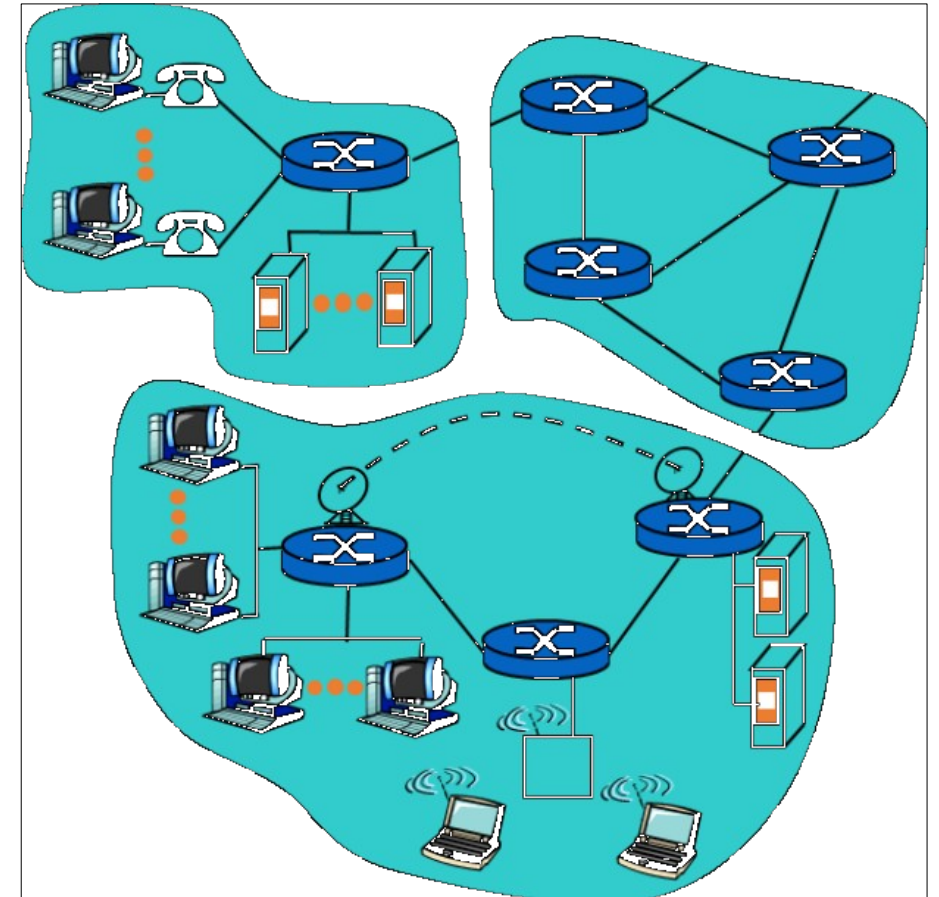


Internet

Internet is a global network of interconnected networks that use standardized protocols to provide common services.

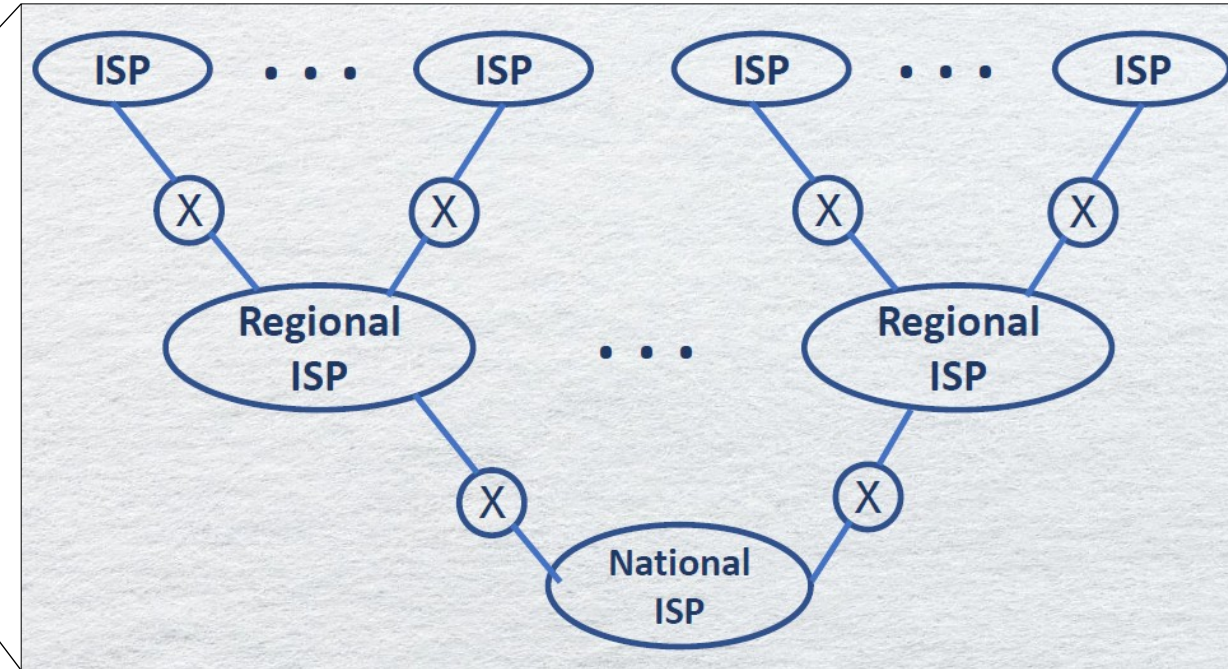
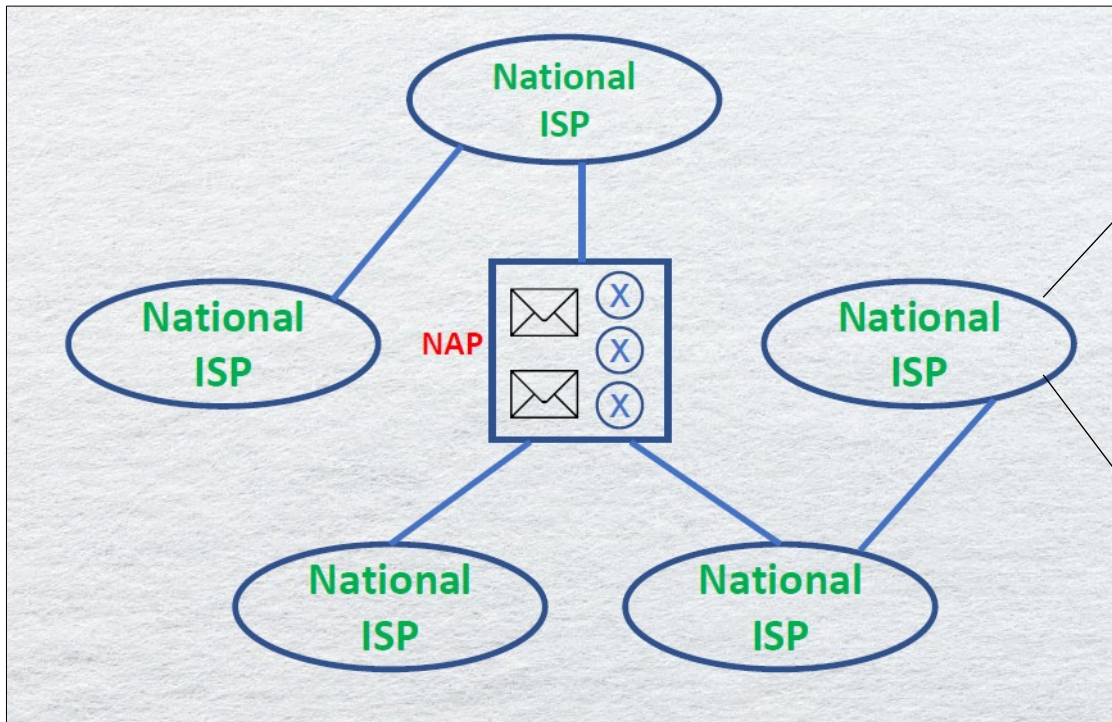
Enables seamless communication across different technological infrastructures.

Connects millions of networks and billions of devices globally.



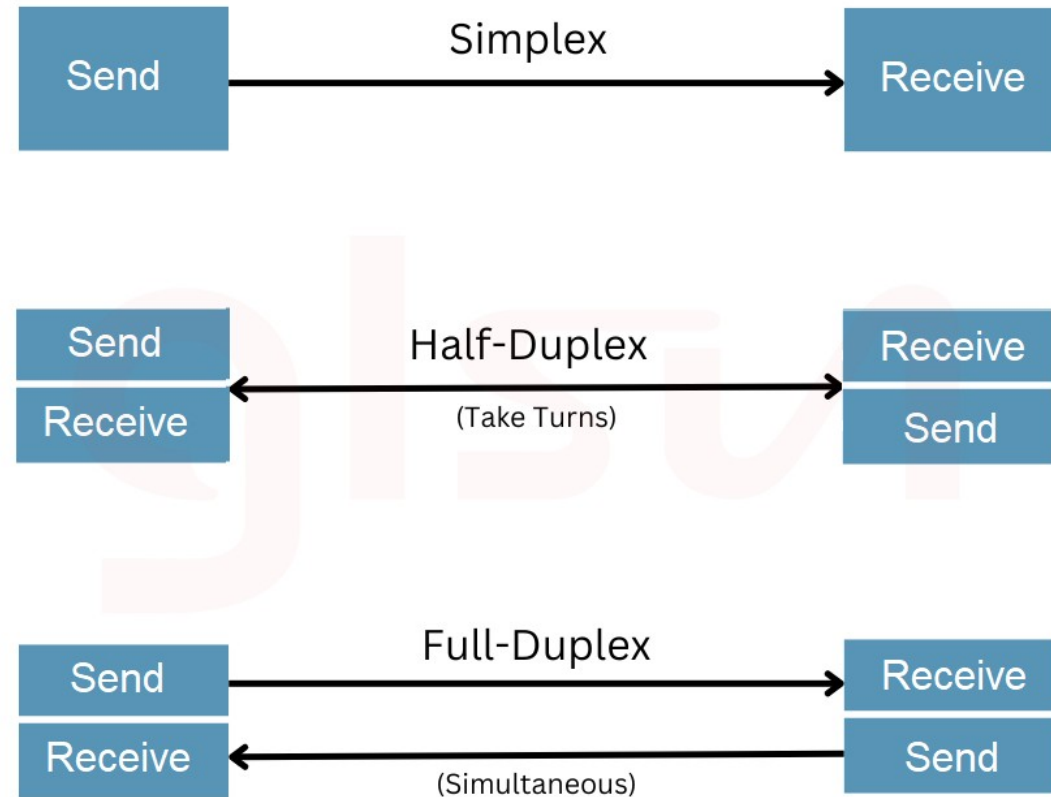
Hierarchical organization of the Internet

NAP: Network Access Point



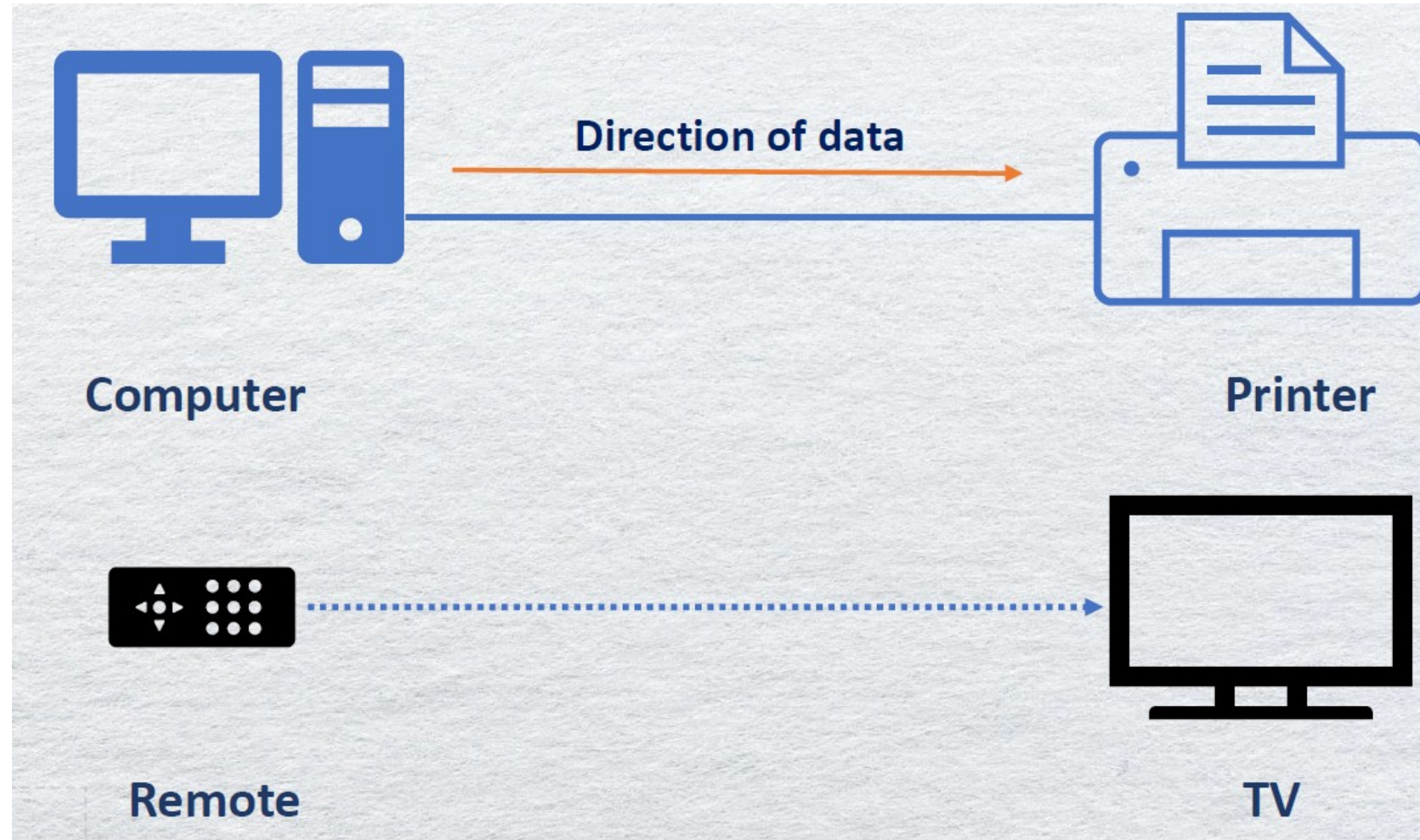
Transmission Modes

Transmission modes define the direction in which data can be transmitted between two connected devices. Communication between two devices can be simplex, half-duplex, or full-duplex.



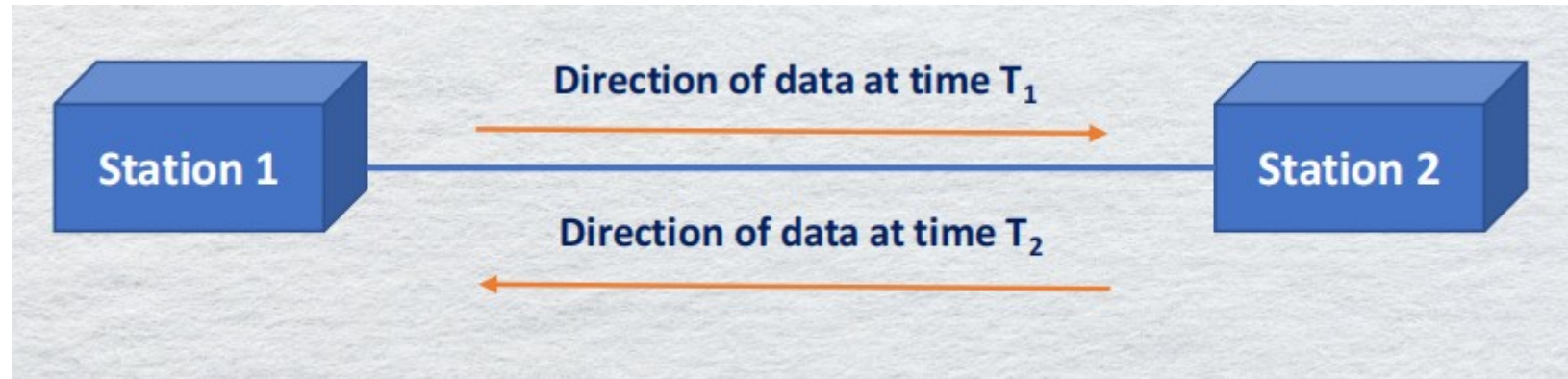
Simplex

- ✓ Unidirectional Communication
- ✓ Sender can only send, and receiver can only receive the data.
- ✓ *E.g.*, keyboard, monitor, television remote, etc.



Half-Duplex

- ✓ Bi-directional communication.
- ✓ Both nodes can send/receive but not at the same time, i.e., if one node is
- ✓ sending other will be receiving and vice versa.
- ✓ E.g., Walkie-talkie.



Duplex/Full Duplex

- ✓ Bi-directional communication.
- ✓ Both nodes can send/receive at the same time.
- ✓ The full-duplex transmission provides the best performance among the three and
- ✓ It maximally utilizes the available bandwidth.
- ✓ E.g., Telephonic communication

